

Chicken Food Collector

Project Overview:

This project proposes the development of a basic Chicken Food Collector Game using Java with a graphical user interface (GUI). The game allows the player to control a chicken and collect corn placed at random positions on the screen. Each time the chicken collects a corn, the score increases and a new corn appears at another random location.

The project is intended as a beginner-friendly application to practice core Java programming concepts such as object-oriented programming, user input handling, GUI development, game logic implementation, and basic problem solving. The focus of the project is on correct functionality and simple game mechanics rather than advanced graphics or complex features.

Objectives:

The objectives of this project are:

- To develop a basic Chicken Food Collector Game using Java.
- To design a simple GUI-based game interface.
- To implement keyboard controls for chicken movement.
- To implement the main game features:
 - Chicken movement using keyboard keys
 - Random corn generation on the game area
 - Score counting when corn is collected
 - Collision detection between chicken and corn
 - Winning condition when the target score is reached
- To display score and winning message.
- To ensure the game runs properly in NetBeans or Visual Studio Code.
- To improve practical understanding of Java programming fundamentals, GUI programming and simple game logic design.

Java Chicken Food Collector (Without Database)

10-Week Plan

1. Week1: Project Planning & Research
 - Select Chicken Food Collector Game
 - Define project objectives
 - Understand game requirements
 - Create project timeline(10 weeks)
2. Week 2: Game Design & Flowchart
 - Design game layout
 - Decide chicken and corn positions
 - Create game flowchart
 - Prepare basic game logic
3. Week 3: Project Setup & GUI
 - Create Java project
 - Create game window
 - Create required classes
 - Test project execution
4. Week4: Chicken Character Development
 - Add chicken character
 - Set starting position
 - Display chicken on screen
5. Week 5: Chicken Movement
 - Add keyboard controls
 - Move chicken using arrow keys
 - Prevent movement outside boundaries
6. Week 6: Corn Generation
 - Add corn object
 - Display corn on screen
 - Set corn position
7. Week 7: Corn Collection & Score
 - Detect chicken and corn collision
 - Increase score after collection
 - Display score on screen
8. Week 8: Random Corn Placement
 - Generate corn at random positions
 - Update score continuously
 - Improve game functionality
9. Week 9: Winning Condition
 - Set target score (10 points)
 - Display winning message

- Complete game logic

10. Week10: Testing & Documentation

- Test all game features
- Fix bugs and errors
- Take screenshots
- Prepare final report and presentation